

## No. of Functional Features Included in the Solution

|               |                                                             |
|---------------|-------------------------------------------------------------|
| Date          | 3 July 2026                                                 |
| Project Name  | Rising Waters - Real-time Flood Prediction and Alert System |
| Maximum Marks | 5 Marks                                                     |

### Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

| S.No | Feature Name             | Feature Description                                                                                | Module / Component    | Status (Done / In Progress / Pending) | Marks Contribution |
|------|--------------------------|----------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------|--------------------|
| 1    | User Registration & Auth | Secure creation and verification of user accounts with credential encryption.                      | Authentication Module | Done                                  | 0.5                |
| 2    | Live Data Ingestion      | Automated REST API collection pipeline gathering live hydrological telemetry metrics.              | Data Pipeline         | Done                                  | 0.5                |
| 3    | Risk Predictive Model    | Random Forest classifier analyzing historical and live stats to compute immediate overflow scales. | ML Engine             | Done                                  | 1.0                |
| 4    | Interactive Map UI       | Geographical tracking board mapping flood gauges and regional warning states.                      | Frontend View         | Done                                  | 1.0                |
| 5    | SMS Broadcast System     | Instant priority notification dispatches initiated when risk levels surpass thresholds.            | Alert Engine          | Done                                  | 0.5                |
| 6    | Historical Trends Charts | Data charts illustrating metric variations for comparative regional monitoring.                    | Analytics Board       | Done                                  | 0.5                |
| 7    | Fail-Safe Local Logs     | Local caching of telemetry inputs during upstream database interruptions.                          | Storage Layer         | Done                                  | 0.5                |
| 8    | Profile Alerts Adjuster  | Preference dashboard letting users customize alert triggers and zone configurations.               | User Settings         | Done                                  | 0.5                |

**Feature Summary:**

| Metric                             | Count / Value |
|------------------------------------|---------------|
| Total Features Planned             | 8             |
| Total Features Implemented         | 8             |
| Core / Must-Have Features          | 5             |
| Additional / Nice-to-Have Features | 3             |
| Features Tested & Verified         | 8             |

**Feature Category Breakdown:**

| S.No | Category                  | Features in Category | Example Features                                    |
|------|---------------------------|----------------------|-----------------------------------------------------|
| 1    | User Interface (UI)       | 3                    | Interactive Map UI, Trend Charts, Profile Dashboard |
| 2    | Backend / Logic           | 2                    | Risk Predictive Model, Live Ingestion Workers       |
| 3    | Database / Storage        | 1                    | Fail-Safe Local Logs Caching                        |
| 4    | API / Integration         | 1                    | SMS Broadcast System (Twilio Integration)           |
| 5    | Security / Authentication | 1                    | User Registration & Auth Management                 |